

Literature Status: Locally L^0 -Convex Topologies

Run fa_banach_001, agent agent_lane_11

17 August 2026

Status. literature_already_answered. Zapata [2] gives one counterexample that answers negatively the cluster of questions following Propositions 2.22–2.24 of Guo–Zhao–Zeng [1]. We spell out two immediate consequences of the example—its gauge is identically zero, and no equivalent local base can consist of countable-concatenation-stable sets—because together they match the source questions exactly.

The questions and the later criterion. For an L^0 -balanced, absorbent, and convex set U in an L^0 -module, write

$$p_U(X) = \text{ess inf}\{\lambda \in L_{++}^0 : X \in \lambda U\}.$$

The source asks whether

$$\{X : p_U(X) < 1\} \subset U \tag{1}$$

can fail without the countable concatenation property, whether every locally L^0 -convex topology is induced by L^0 -seminorms, and whether the latter conclusion can be rescued when the ambient module itself has the countable concatenation property.

Zapata’s characterization says that a topology on an L^0 -module is induced by a family of L^0 -seminorms if and only if it has a zero-neighborhood base of L^0 -balanced, absorbent, convex sets that are closed under countable concatenations. His following example shows that the extra condition is essential.

The counterexample. Let $(A_n)_{n \geq 1}$ be an infinite measurable partition of Ω with $P(A_n) > 0$, let $E = L^0(\Omega, \mathcal{F}, P)$, and for $\varepsilon \in L_{++}^0$ set

$$U_\varepsilon = \{Y \in L^0 : \exists I \subset \mathbb{N} \text{ finite such that } |Y \mathbf{1}_{A_i}| \leq \varepsilon \text{ for every } i \notin I\}. \tag{2}$$

Zapata verifies that the family (U_ε) is a zero-neighborhood base for a locally L^0 -convex module topology. The elementary checks are transparent: finite exceptional sets are preserved under balanced scaling and finite unions handle convex combinations; absorbency follows since

$$\lambda = 1 + |Y|/\varepsilon \in L_{++}^0 \implies |Y/\lambda| \leq \varepsilon.$$

The ambient module $E = L^0$ has the countable concatenation property.

On the other hand, every U_ε is proper because $\varepsilon + 1 \notin U_\varepsilon$. Yet its countable-concatenation hull is all of L^0 :

$$H_{cc}(U_\varepsilon) = L^0. \tag{3}$$

Indeed, for any $Y \in L^0$, each one-piece truncation $Y \mathbf{1}_{A_n}$ belongs to U_ε , and their concatenation along (A_n) is Y .

No countable-concatenation local base. Suppose an equivalent zero-neighborhood base contained a set V closed under countable concatenations with $V \subset U_\varepsilon$. Because V is a neighborhood in the topology generated by (2), some $U_\delta \subset V$. Then (3) and concatenation stability give

$$L^0 = H_{cc}(U_\delta) \subset V \subset U_\varepsilon,$$

contradicting the properness of U_ε . Thus even though E has the countable concatenation property, its topology need not have the kind of local base proposed in [1]. By Zapata’s criterion, this topology is not induced by any family of L^0 -seminorms.

The gauge is identically zero. The same example gives a sharp failure of (1). Fix $X \in L^0$, an index m , and a real $t > 0$. Define

$$\lambda_{m,t} = t\mathbf{1}_{A_m} + \left(1 + \frac{|X|}{\varepsilon}\right) \mathbf{1}_{\Omega \setminus A_m} \in L_{++}^0.$$

Outside the single exceptional piece A_m , $|X/\lambda_{m,t}| \leq \varepsilon$, so $X/\lambda_{m,t} \in U_\varepsilon$ and hence $p_{U_\varepsilon}(X) \leq \lambda_{m,t}$. Restricting to A_m and then letting $t \downarrow 0$ gives $p_{U_\varepsilon}(X) = 0$ on A_m . Since the A_m partition Ω ,

$$p_{U_\varepsilon}(X) = 0 \quad \text{for every } X \in L^0. \quad (4)$$

Consequently

$$\{X : p_{U_\varepsilon}(X) < 1\} = L^0 \not\subset U_\varepsilon,$$

which is the requested explicit negative answer to the random-gauge question.

Independent confirmation and scope. Guo–Zhao–Zeng later explicitly record that Wu–Guo and Zapata independently supplied counterexamples to seminorm generation [3]. The result above resolves the source’s topology/gauge question cluster. It does *not* claim to resolve the distinct open problem later in [1]: whether an L^0 -pre-barreled module with the countable concatenation property must be L^0 -barreled. A bounded search found no later resolution of that implication, so it is excluded from the present result.

References

- [1] T. Guo, S. Zhao, and X. Zeng, *On random convex analysis—the analytic foundation of the module approach to conditional risk measures*, arXiv:1210.1848v3.
- [2] J. M. Zapata, *On characterization of locally L^0 -convex topologies induced by a family of L^0 -seminorms*, arXiv:1404.0357v4.
- [3] T. Guo, S. Zhao, and X. Zeng, *Random convex analysis (I): separation and Fenchel–Moreau duality in random locally convex modules*, arXiv:1503.08695v2.